Problem-Solution Fit Canvas

Date	14 July 2026
Team ID	
Project Name	HDI Predictor (Human Development Index Predictor)
Team Size	1 (Individual Project)
Maximum Marks	5 Marks

1. Customer Segment(s) [CS]

Policymakers, development researchers, and students who need to interpret or estimate a country's Human Development Index without manually working through UNDP formulas.

2. Problems / Pains + Frequency [PR]

Recomputing HDI by hand for hypothetical scenarios is slow and error-prone; happens every time a new policy scenario needs evaluating (frequent, recurring).

3. Triggers to Act [TR]

A report deadline, a classroom assignment, or a policy meeting requiring a quick answer to “what tier would this country fall into?”

4. Emotions Before / After [EM]

Before: frustrated, uncertain about accuracy. After: confident, because the prediction is backed by a model trained on real UNDP data with a stated R^2.

5. Available Solutions Pros & Cons [AS]

Static UNDP PDF tables (accurate but not interactive); manual spreadsheets (flexible but formula-heavy and error-prone); no existing lightweight web tool focused specifically on this prediction task.

6. Customer Limitations [CL]

Limited time to learn the HDI formula in depth; may not have Python/Excel modeling skills; needs results on any device (desktop or mobile).

7. Behavior + Its Intensity [BE]

Repeatedly re-checks a handful of scenarios in a single working session (high-frequency, low-effort-per-check behavior).

8. Channels of Behavior [CH]

Online — accessed as a web app from any browser, no installation.

9. Problem Root / Cause [RC]

HDI's official formula (log-income sub-index + geometric mean across three dimensions) is not something most users can quickly compute by hand.

10. Your Solution [SL]

A Flask web app with a model trained on real UNDP data ($R^2 = 0.999$) that instantly predicts HDI and its tier from four indicators, with results logged for later review.